Dashboard Guide: Every Chart and Result

Breathing Cities, DWDM (CSI3010). How to read and explain each part of the Streamlit dashboard. Numbers are from the real run with default settings.

Starting the dashboard

In VS Code, open the `breathing-cities-dwdm` folder, open a terminal (View → Terminal), then run:

`.venv\Scripts\Activate.ps1` then `streamlit run dashboard/app.py`

It opens in the browser at `http://localhost:8501`. No internet is needed except for the map background.

The layout

Part	What it is
Header banner	Project title, one-line summary and key facts: 26 cities, 110 stations, 29,531 city-days, 3 data sources, a fact constellation of 9 tables.
Sidebar (left)	**Cities to compare** (up to 4; default Delhi, Bengaluru, Mumbai) and a **Years** slider. These filter the Overview tab. *Say: "The sidebar is an interactive slice and dice on the warehouse."*
7 tabs	Overview, OLAP explorer, Stations & weather, Prediction, Patterns, Lockdown what-if, Diwali.
Blue boxes with a bulb	The one-line takeaway for that chart. You can read them aloud.

Tab 1: Overview

Five KPI cards (headline numbers)

Card	Default value	Meaning
City-days	6,027	Daily records matching the sidebar filters.
Average AQI	166, Moderate	Average air-quality score (0-500, higher is worse) and its official bucket.
Unhealthy days	23%	Share of days with AQI above 200 (Poor or worse): about 1 day in 4.
Worst selected city	Delhi (AQI 258)	Highest average among the selected cities (Mumbai 105, Bengaluru 94).
Lockdown effect	-28%	PM2.5 during lockdown compared with what the weather model expected (see Tab 6).

India map

Each bubble is a city. **Bigger and darker orange = worse average AQI.** North India is dark: Delhi (258), Patna (240), Gurugram (224). The north-east and south are pale: Aizawl (35), Shillong (54), Coimbatore (73).

Average AQI by city (bar chart)

All 26 cities ranked, worst at the top. Your selected cities are orange, the rest grey.

Say: "Pollution is concentrated in the Indo-Gangetic plain in the north: landlocked, cold still winters, crop burning and heavy traffic. Coastal and hill cities are far cleaner."

Monthly AQI line chart

One line per selected city, 2015-2020. The grey band marks the 2020 lockdown. Point out the **yearly wave** (high in Nov-Jan, low in Jul-Sep because rain washes pollution out), Delhi's line far above the others, and the drop inside the lockdown band.

Seasonal fingerprint (heatmap)

Rows are cities (worst at the top), columns are months, colour is average AQI (darker = worse). Read it like a calendar: the north Indian rows are dark from October to February, and almost every row is pale in July-September. Grey cells mean no data for that city and month.

Tab 2: OLAP explorer

Choose one of 9 operations with the buttons. For each, the dashboard runs the real SQL against the warehouse and shows three things: a **chart** (left), the **result table** (right) and the **SQL** (right, below).

Say: "This runs the actual SQL on the data warehouse, live."

Operation	Chart	Key result
Roll-up (zoom out: city → region)	One line per region, average PM2.5 per year	North: 130 (2016) → 79 (2019) → 55 (2020; half-year incl. lockdown)
Drill-down (zoom in: year → month)	Delhi 2019 month by month; bar colour = official AQI bucket	AQI 364 in January, 99 in August, 341 in November
Slice (fix one value: Winter)	Winter AQI per city, coloured by bucket	Delhi 330, Patna 328, Lucknow 307
Dice (several filters: North x Mar-May x 2019/2020)	Grey = 2019, blue = 2020, per city	Delhi PM2.5 85 → 53; every northern city cleaner in 2020
Pivot (buckets become columns)	Each bar = 100% of a city's days, split by AQI bucket colour (green = Good ... dark red = Severe)	Delhi: 239 Severe and 520 Very Poor days. More red = worse city.
Event dimension	Average AQI per event	Stubble burning 293, Diwali 218, Normal 149, Lockdown 88

Drill-down to stations	Delhi's 38 stations, Nov 2019, sorted	PM2.5 from 148 to 252: a 1.7x difference within one city
Dice on weather	Heatmap: wind (rows) x mixing height (columns), % of winter days unhealthy	Calm + low mixing: 60%; breezy + high mixing: 1%
Drill-across	Two stacked panels sharing the month axis: PM2.5 (from the pollution fact) and mixing height (from the weather fact)	Delhi's worst months (Nov-Jan, PM2.5 ~205) are the months with the lowest mixing height (260-370 m)

Mixing height = how high the air can mix, like the height of a ceiling over the city. Low ceiling = pollution trapped near the ground. **Drill-across** = combining two fact tables through the shared Date and City dimensions; it is possible because the warehouse is a fact constellation.

Tab 3: Stations & weather

Station heatmap

Pick a city. Rows are its monitoring stations, columns are years, each cell is average PM2.5 with the number written in. Stations are sorted by their 2019 level, worst first. Blank cells mean the station was not reporting yet (most Delhi stations opened in 2018). In Delhi, Wazirpur, Bawana and Anand Vihar are worst; most rows get lighter towards 2019-2020.

Say: "This drills below city level using the station fact table. The same city has neighbourhoods with very different air."

Line chart: winter smog fades as the mixing height rises

X axis = mixing height (log scale), Y axis = average PM2.5, one line per season. Read it carefully:

- **Winter (blue) and post-monsoon (yellow):** PM2.5 roughly halves, from about 125-140 down to about 60, as mixing height rises from 200 m to 1,000 m.
- **Pre-monsoon (orange) and monsoon (green):** already low (35-65) and fairly flat. Rain and sea breezes clean the air, so the ceiling matters little. They rise slightly at very high mixing heights (hot, windy, dusty days).

Heatmap: % of winter days unhealthy, by wind x mixing

Wind \ Mixing	Low	Medium	High
Calm	60%	39%	no data
Light	58%	21%	12%
Breezy	54%	22%	1%

Same season, very different air. The **Low** column is always above 50%, so mixing height matters more than wind.

Tab 4: Prediction

Classification: will tomorrow be unhealthy?

Trained on 2015-2018, tested on 2019-2020 (train on the past, test on the future).

Card	Value	Meaning
Random Forest F1	0.83	Main score (0-1, higher is better); balances precision and recall.
Baseline F1	0.78	"Tomorrow = today". The model must beat this.
Change days predicted	35%	Only 6.6% of days switch between healthy and unhealthy, the days that matter most. The baseline gets 0% of them by definition.
ROC-AUC	0.981	How well risky days are ranked above safe ones (0.5 = guessing, 1 = perfect).

Model comparison chart: blue = F1, orange = accuracy on change days, for Baseline (0.78 / 0%), Decision Tree (0.81 / 28%), Random Forest (0.83 / 35%), Random Forest + weather (0.83 / 35%).

SHAP chart: longer bar = that input mattered more. Top: today's AQI, today's PM2.5, 3-day average AQI, then PM10, CO, North region, NOx, rain. The model relies on recent pollution; pollution builds up over days.

Say: "Random Forest beats the baseline and predicts a third of the days when air quality changes. SHAP explains why, so it is not a black box. Weather did not add accuracy for next-day prediction, because today's pollution already reflects today's weather."

Forecasting monthly AQI for 2019

MAPE chart (average % error, lower is better; 7 cities with 3+ years of history): seasonal naive 16.8%, Holt-Winters 23.9%, SARIMA 24.2%, **weather-adjusted 14.2%** (blue, best). Weather-adjusted wins in 12 of 14 cities.

City line chart (pick a city): black = actual 2019 AQI, dotted grey = seasonal naive (same month last year), blue = weather-adjusted. The closer to black, the better. For Delhi the blue line tracks the U-shape most closely.

Say: "Most cities have only 1-3 years of history, too little for Holt-Winters or SARIMA. Adjusting last year's value for this year's weather was most accurate."

Tab 5: Patterns

Clustering: city pollution profiles

Scatter: each dot is a city, coloured by K-Means cluster. The axes come from **PCA**, which squeezes each city's 9 numbers into 2 so they can be drawn. The axis values have no unit: **only distance matters; nearby cities are similar.** Delhi sits far right on its own (most extreme). Hover over a dot to see its numbers.

Cluster (card)	Cities	Profile

Smog belt (orange)	Delhi, Gurugram, Lucknow, Patna	PM2.5 90, NO2 31, unhealthy 50% of days; hit hardest by low mixing height
Moderate urban (blue)	16 cities incl. Mumbai, Bengaluru, Chennai, Kolkata	PM2.5 38, traffic NO2 21, unhealthy 7%
Low-traffic, SO2-heavy (green)	Aizawl, Ernakulam, Guwahati, Jorapokhar, Kochi, Talcher	Very little NO2 (7) but more SO2 (16, coal and industry); unhealthy 11%

k = 3 was chosen by the silhouette score (k=2: 0.227, k=3: 0.226, a tie, so the more detailed answer was kept).

Association rules: which conditions bring bad air?

Pick a region. Each bar is a rule ("if these conditions..."). **Bar length** = how often the air is unhealthy when the conditions hold. The **orange dashed line** = the region's normal rate. The bar label shows the percentage and the **lift** (how many times more likely than normal; 2.0 = twice as likely).

Region	Top rule	Unhealthy	Normally	Lift
North	stubble burning + low mixing + mild	83%	39%	2.0
East	low mixing + dry + cold	81%	26%	2.3
North-East	dry + winter + mild	91%	14%	5.5

The **Co-pollution rules** drop-down shows which pollutants exceed their limits together (e.g. NO2 high → PM2.5 high, 86% in the East): pollutants rising together suggest a shared source such as traffic.

Careful: Central (only Bhopal) and West have few unhealthy days, so their rules rest on small numbers. Lead with North, East and North-East.

Tab 6: Lockdown what-if (end the demo here)

Three KPI cards

Pollutant	Lockdown 2020	Placebo 2019
NO2 (traffic gas)	**-29%**	-1%
PM2.5	**-28%**	-4%
AQI	-19%	-4%

Each value is the median city: actual pollution (25 Mar - 31 May 2020) compared with what a weather model predicted without a lockdown. The **placebo** repeats the test on 2019, when there was no lockdown; it gives almost 0, which proves the method does not invent effects.

Delhi 2020 line chart

Blue = actual daily PM2.5, dotted grey = expected without lockdown, grey band = lockdown. Before the lockdown the lines match (January: actual 157, expected 150), so the model works. Inside the band, blue sits well below grey (April: 45 vs 68). That gap is the lockdown effect.

Placebo chart and city chart

Left: the KPI numbers as bars; blue (2020) long, grey (2019) tiny. Right (pick a pollutant): one bar per city; **blue = cleaner than expected, red = dirtier**. PM2.5 fell most in Aizawl and Ahmedabad (-52%) and Chennai (-48%); Delhi -28%. Coal and mining towns Brajrajnagar (+28%) and Jorapokhar (+23%) got dirtier because power plants and mines kept running.

Careful (NO2 view): two extreme cities (Coimbatore and Aizawl) are hidden because their NO2 levels are so low that tiny changes become huge percentages. That is why headline numbers use the median city.

Say: "After removing the weather's effect, the lockdown cut NO2 by 29% and PM2.5 by 28%. The same test on a normal year shows almost nothing, so the effect is real."

Tab 7: Diwali

Three KPI cards

395 → 301: Delhi's raw Diwali PM2.5 fell from 2016 to 2017, so the 2017 firecracker sales ban looks like it worked. **+126%**: but 2017's weather was favourable (expected only 134 without fireworks), so the firework spike was actually the largest. **+45%**: the spike in 2018-19 under the green-cracker rules (8-10 pm only).

Delhi bar chart

Per year, grey = expected PM2.5 from weather alone, orange = actual; the label on top is the extra % from fireworks: 2015 +36%, 2016 +63%, **2017 +126%** (sales ban), 2018 +47% (green crackers), 2019 +45%.

All-cities chart

Each dot is one city's Diwali spike in that year; the black line joins the median city: 2015 +42%, 2016 +67%, 2017 +44%, 2018 +32%, 2019 +44%. Diwali typically adds 30-70% extra PM2.5. Dots below 0 are cities where rain or wind cleared the air that night. More dots in later years because more cities started reporting.

Say: "Raw numbers made the 2017 ban look effective. Once we remove the weather's effect, the firework spike was actually the largest that year. The weather made the ban look good."

Suggested demo order (about 4 minutes)

1. **Overview:** the map and ranking (north vs south), then the KPI cards.
2. **OLAP explorer:** Roll-up, then Event dimension, then Drill-across.
3. **Stations & weather:** the Delhi station heatmap and the wind x mixing heatmap.
4. **Prediction:** the KPI cards and the SHAP chart.
5. **Patterns:** the three clusters, then the North rule (83% vs 39%).
6. **Lockdown what-if:** the Delhi 2020 chart. Strongest finish.
7. **Diwali:** the 2017 story, if time allows.